JolSetu (■■■■■■) — the people's bridge to safe water

Xylem Global Student Innovation Challenge 2026 · Water Quality · University track · rudra496.github.io/jolsetu · github.com/rudra496/jolsetu

The problem

Bangladesh's groundwater arsenic is “the largest poisoning of a population in history” (WHO Bulletin 2000). **20 million** people drink water above the national 50 µg/L standard, **45 million** above the WHO 10 µg/L guideline; arsenic causes **24,000+ adult deaths a year** and **US$13B** in productivity losses over 20 years (Flanagan et al., *Bull WHO* 2012). Ten million wells were screened once in national campaigns — but there is no live map, readings are human color-chart guesses, and families have no way to find the nearest verified safe well.

The solution — four moves, one bridge

1 · Read — a phone camera + color science (white-reference normalization, CIELAB matching) reads any $1 Gutzeit kit instantly and speaks the verdict in Bangla (approach validated: Haque et al., *Sci Total Environ* 2018). **2 · Stamp** — every test is HMAC-signed, GPS-geofenced (30 m), duplicate-checked: a tamper-proof public map. **3 · Learn** — a disclosed risk model (AUC 0.67 on 1,319 held-out real wells) finds where map and model disagree; the crowd tests there next — an active-learning loop. **4 · Walk** — distance, bearing and walking time to the nearest verified safe wells, offline, in Bangla.

20M	52.9%	76% → 11%	99.3 / 63%	23/23
exposed above the national standard (WHO 2012)	of the 6,593 real wells in our live pilot dataset are unsafe	unsafe share: 50–100 ft wells vs wells below 150 ft — drill deep	human kit reading sensitivity / specificity (n=943) — what AI fixes	automated tests passing; PWA installs offline (BN+EN)

Why it can win

Impact — well-switching at ≈US$11 per person (UNICEF program, Bull WHO 2012) immediately cuts exposure; the same stack covers the 94M+ people of the global arsenic belt (Podgorski & Berg, *Science* 2020). **Challenge fit** — detect + monitor + prevent arsenic, the largest single water-contaminant health burden; no new hardware: the kit already exists in every bazaar. **Innovation** — first CV reader for the existing $1 kit, first active-learning testing queue, first fraud-proof crowd map (vs iArsenic's static 3-input screen, Nolkup's registry, Arsenic 4M's new hardware).

Built, running, honest

The prototype is live on GitHub Pages, built on 6,593 real Araihazar measurements (HydroShare DOI 10.4211/hs.8e1373d87419447c945625af13f0a2ea) with a published model card, claim→source evidence file, and 23 passing tests. 8-week plan: 2-school pilot with per-batch calibration → server ledger with officer review → national model v1 → SMS fallback → published impact audit (exposure-days avoided).

Every DOI verified live via Crossref/PubMed. Full evidence map: docs/EVIDENCE.md in the repository. Team: Rudra Sarker (+up to 4).